



Merewether High School

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2025 **Trial HSC Examination**

Year 12 Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes.
- Working time – 3 hours.
- Write using black pen.
- NESA approved calculators may be used.
- A reference sheet is provided.
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- Submit ALL work at the end of the examination.
- Diagrams are NOT drawn to scale.

**Total marks:
100**

Section I – 10 marks (pages 2 – 6)

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.

Section II – 90 marks (pages 7 – 15)

- Attempt Questions 11 – 16.
- Allow about 2 hour and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 Consider the statement

‘for any integers m and n , if $m + n \geq 9$ then $m \geq 5$ or $n \geq 5$ ’.

The contrapositive of this statement is:

- A. If $m < 5$ or $n < 5$, then $m + n < 9$
 - B. If $m \geq 5$ or $n \geq 5$, then $m + n \geq 9$
 - C. If $m < 5$ and $n < 5$, then $m + n < 9$
 - D. If $m \leq 5$ and $n \leq 5$, then $m + n \leq 9$
- 2 A particle starts from rest, 2 metres to the right of the origin, and moves along the x – axis in simple harmonic motion with a period of 2 seconds.

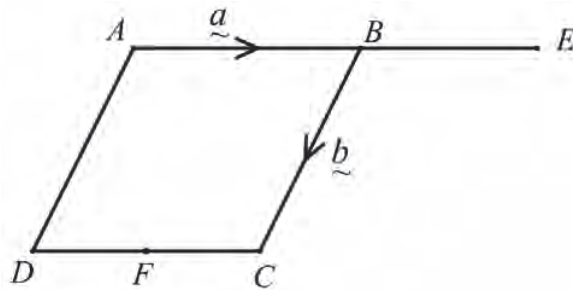
Which equation could represent the motion of the particle?

- A. $x = 2 \cos \pi t$
- B. $x = 2 \cos 2t$
- C. $x = 2 + 2 \sin \pi t$
- D. $x = 2 + 2 \sin 2t$

- 3 If $z = -(2a+1) + 2ai$, where a is a non-zero constant, then $\frac{4a}{1+\bar{z}}$ is equal to:

- A. $\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$
 B. $\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$
 C. $\sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$
 D. $\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)$

- 4 $ABCD$ is a rhombus. $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$.



$\overrightarrow{BE}$ is an extension of $\overrightarrow{AB}$, such that $\overrightarrow{AB} : \overrightarrow{BE} = 4 : 3$.

The point F is the midpoint of CD .

Which expression gives the vector $\overrightarrow{FE}$ in terms of $\underline{a}$ and $\underline{b}$?

- A. $\frac{5}{4}\underline{a} - \underline{b}$
 B. $\frac{5}{4}\underline{a} + \underline{b}$
 C. $\frac{11}{6}\underline{a} - \underline{b}$
 D. $\frac{11}{6}\underline{a} + \underline{b}$

5 Using the substitution $u = 1 + e^x$, $\int_0^{\log_e 2} \frac{1}{1 + e^x} dx$ can be expressed as:

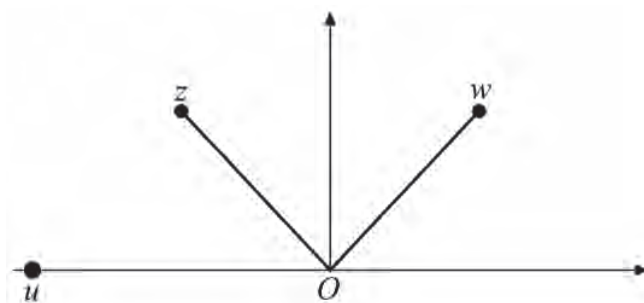
A. $\int_0^{\log_e 2} \left(\frac{1}{u-1} - \frac{1}{u} \right) du$

B. $\int_2^3 \left(\frac{1}{u} - \frac{1}{u-1} \right) du$

C. $\int_1^3 \left(\frac{1}{u} - \frac{1}{u-1} \right) du$

D. $\int_2^3 \left(\frac{1}{u-1} - \frac{1}{u} \right) du$

6 The Argand diagram shows the complex numbers w , z and u , where w lies in the first quadrant, z lies in the second quadrant and u lies on the negative real axis.



Which statement could be true?

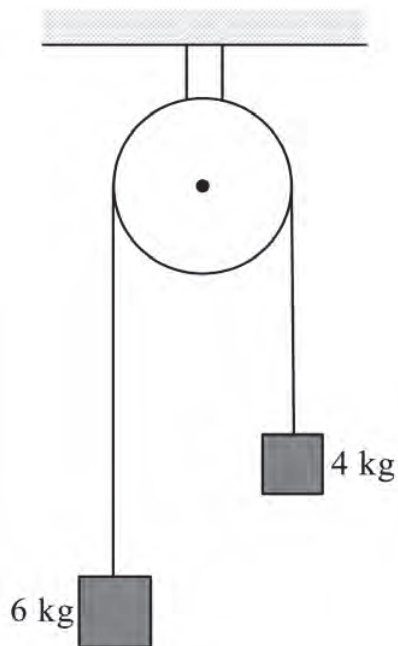
A. $u = zw$ and $u = z + w$

B. $u = zw$ and $u = z - w$

C. $z = uw$ and $u = z + w$

D. $z = uw$ and $u = z - w$

- 7 A light string passes over a smooth pulley. Attached to the ends of the string are masses of 6 kg and 4 kg, as shown.



The acceleration due to gravity is $g \text{ ms}^{-2}$.

What is the acceleration of the 6 kg mass?

- A. $\frac{1}{5}g$
- B. $\frac{1}{2}g$
- C. $\frac{2}{5}g$
- D. g
- 8 The complex number z satisfies $|z - 1| = 1$.
- What is the greatest distance that z can be from the point i on the Argand diagram?
- A. 1
- B. $\sqrt{5}$
- C. $2\sqrt{2}$
- D. $\sqrt{2} + 1$

- 9 Consider the line segments given by the vector equations below.

$$\vec{r} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \end{pmatrix}, \quad -2 \leq \lambda \leq 2;$$

$$\vec{l} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad 0 \leq \mu \leq 2;$$

$$\vec{v} = \gamma \begin{pmatrix} 9 \\ -6 \end{pmatrix}, \quad 1 \leq \gamma \leq 3$$

Which of the following statements about the line segments is NOT true?

- A. At least two of them are parallel
- B. At least two of them are perpendicular
- C. At least two of them have the same intercept
- D. At least two of them have the same length

- 10 Without evaluating the integrals, which one of the following integrals is greater than zero?

A. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{2 + \cos x} dx$

B. $\int_{-\pi}^{\pi} x^3 \sin x dx$

C. $\int_{-1}^1 (e^{-x^2} - 1) dx$

D. $\int_{-2}^2 \tan^{-1}(x^3) dx$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Points A , B and C in three-dimensional space have position vectors:

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}.$$

(i) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$. **1**

(ii) Hence, find the acute angle between $\overrightarrow{AB}$ and $\overrightarrow{AC}$, **2**
correct to the nearest degree.

(b) Find $\int \frac{1}{x^2 + 2x + 3} dx$. **2**

(c) Consider the complex number $z = (b - i)^3$, where $b \in \mathbb{R}^+$.
Find b given that $\arg(z) = -\frac{\pi}{2}$. **3**

(d) Sketch the region in the Argand diagram defined by **2**
 $6 \leq \operatorname{Re}[(2 - 3i)z] < 12$ and $\operatorname{Re}(z)\operatorname{Im}(z) \geq 0$.

Question 11 continues page 8

Question 11 (continued)

- (e) (i) Find A , B and C such that: **3**

$$\frac{x}{(x-2)^2(x-1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x-1}.$$

- (ii) Hence show that: **2**

$$\int_3^4 \frac{x}{(x-2)^2(x-1)} dx = \ln\left(\frac{3}{4}\right) + 1.$$

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Prove that, if n is an even integer, then $n^3 - n$ is divisible by 6. 2

- (b) (i) Show that the derivative of $\sec \theta$ is $\sec \theta \tan \theta$. 1

- (ii) Use the substitution $x = \sec \theta - 1$ to find the exact value of 4

$$\int_{\sqrt{2}-1}^1 \frac{2}{(x+1)\sqrt{x^2+2x}} dx.$$

- (c) Find a vector perpendicular to both $\underline{u} = -5\underline{i} + 2\underline{j} + 3\underline{k}$ and $\underline{v} = 2\underline{i} - \underline{j} - 2\underline{k}$. 3

- (d) The points P and Q have position vectors:

$$\overrightarrow{OP} = -2\underline{i} + 14\underline{j} - 5\underline{k} \text{ and } \overrightarrow{OQ} = -\underline{i} + 12\underline{j} - 2\underline{k}$$

- (i) Show that the equation of the line l_1 that passes through P and Q is 2

$$\underline{r}_1 = (-2 + \lambda)\underline{i} + (14 - 2\lambda)\underline{j} + (-5 + 3\lambda)\underline{k}$$

where λ is a real number.

- (ii) Consider the line l_2 with the equation 3

$$\underline{r}_2 = (2 + a\mu)\underline{i} + (27 + (a+1)\mu)\underline{j} + (1 + (a+2)\mu)\underline{k}$$

where a is a constant and μ is a real number.

The line l_2 intersects l_1 at the point R . Find the coordinates of R .

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Given that w is a complex cube root of unity, determine the exact value of **2**

$$1 + w + w^2 + w^3 + \dots + w^{1001}.$$

- (b) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$, where n is an integer, and $n \geq 3$.

(i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$. **3**

(ii) Hence evaluate I_7 . **4**

- (c) A particle is moving horizontally. Initially the particle is at the origin O moving with velocity 1 ms^{-1} .

The acceleration of the particle is given by $\ddot{x} = x - 1$, where x is its displacement at time t .

(i) Show that the velocity of the particle is given by $\dot{x} = 1 - x$. **3**

(ii) Find an expression for x as a function of t . **2**

(iii) Find the limiting position of the particle. **1**

End of Question 13

Question 14 (14 marks) Use the Question 14 Writing Booklet

- (a) (i) Let x and y be real numbers such that $x \geq 0$ and $y \geq 0$.

Prove that $\frac{x+y}{2} \geq \sqrt{xy}$. **1**

- (ii) Suppose that a, b, c are real numbers.

Prove that $a^4 + b^4 + c^4 \geq a^2b^2 + a^2c^2 + b^2c^2$. **2**

(iii) Show that $a^2b^2 + a^2c^2 + b^2c^2 \geq a^2bc + b^2ac + c^2ab$. **2**

(iv) Deduce that if $a + b + c = d$, then $a^4 + b^4 + c^4 \geq abcd$. **1**

- (b) A submarine of mass m is travelling underwater at maximum power. At maximum power, its engines deliver a force F on the submarine. The water exerts a resistive force proportional to the square of the submarine's speed v .

- (i) Explain why **1**

$$\frac{dv}{dt} = \frac{1}{m}(F - kv^2)$$

where k is a positive constant.

- (ii) The submarine increases its speed from v_1 to v_2 . Show that the distance travelled during this time is **3**

$$\frac{m}{2k} \log_e \left(\frac{F - kv_1^2}{F - kv_2^2} \right).$$

Question 14 continues page 12

Question 14 (continued)

(c) (i) Prove that $1 - x \leq e^{-x}$ for $x \geq 0$. 2

(ii) $X_1, X_2, X_3, \dots, X_n$ are independent Bernoulli random variables with parameters $p_1, p_2, p_3, \dots, p_n$ respectively. The random variable T is given by $T = X_1 + X_2 + X_3 + \dots + X_n$ and $E(T) = \mu$ denotes the mean of T . Prove that $P(T = 0) \leq e^{-\mu}$. 2

End of Question 14

Question 15 (16 marks) Use the Question 15 Writing Booklet

- (a) (i) Solve $x^2 > 2x + 1$. **1**
- (ii) Prove by mathematical induction that $2^n > n^2$ for all integers $n \geq 5$. **3**
- (b) Prove by contradiction that if x is a rational number ($x \in \mathbb{Q}$), then $x^2 \neq 2$. **3**
- (c) Let $P(z) = z^8 - \frac{5}{2}z^4 + 1$. The complex number w is a root of $P(z) = 0$.
- (i) Show that iw and $\frac{1}{w}$ are also roots of $P(z) = 0$. **2**
- (ii) Find one of the roots of $P(z) = 0$ in exact form. **2**
- (iii) Hence find all the roots of $P(z) = 0$. **2**
- (d) For three 3-dimensional vectors, $\underline{p} = (a, b, c)$, $\underline{q} = (e, f, g)$ and $\underline{r} = (x, y, z)$, **3**
prove that the scalar product is distributive over vector addition,
i.e. prove that:

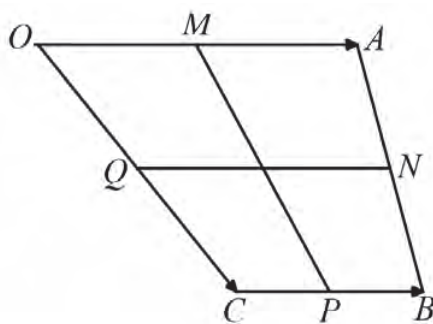
$$\underline{p} \bullet (\underline{q} + \underline{r}) = \underline{p} \bullet \underline{q} + \underline{p} \bullet \underline{r}$$

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) A body of mass m kg is projected from point O with initial speed $V \text{ ms}^{-1}$ at an angle α above the horizontal. It moves in a vertical plane under gravity in a medium which the resistance to motion has magnitude mkv Newtons where $v \text{ ms}^{-1}$ is its speed and $k > 0$ is a constant. What is an expression for the horizontal distance x metres travelled by the body in the first t seconds of its motion? 3

(b)



In the diagram, $OABC$ is a trapezium in which $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$ and $\overrightarrow{CB} = k\overrightarrow{OA}$ for some constant $0 < k < 1$. The points M , N , P and Q are the midpoints of OA , AB , BC and CO respectively.

- (i) Use vector methods to show that MP and QN bisect each other. 2
- (ii) If $MP \perp QN$, use vector methods to show that $|\overrightarrow{OC}|^2 = |\overrightarrow{AB}|^2$. 4

Question 16 continues page 15

Question 16 (continued)

- (c) (i) Use the formula for the sum of a geometric series to show that

$$\sum_{k=1}^n (z + z^2 + \dots + z^k) = \frac{nz}{1-z} - \frac{z^2}{(1-z)^2} (1-z^n), \quad z \neq 1. \quad 2$$

- (ii) Let $z = \cos \theta + i \sin \theta$, where $0 < \theta < 2\pi$.

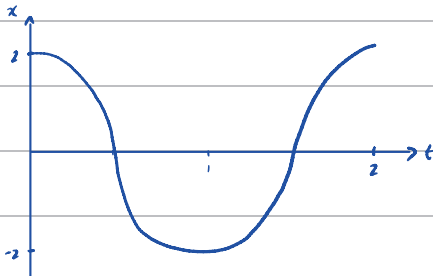
By considering the imaginary part of the left-hand side of the equation of part (i), deduce that

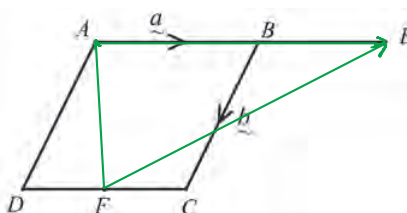
$$\sum_{k=1}^n (\sin \theta + \sin 2\theta + \dots + \sin k\theta) = \frac{(n+1)\sin \theta - \sin(n+1)\theta}{4\sin^2 \frac{\theta}{2}}. \quad 4$$

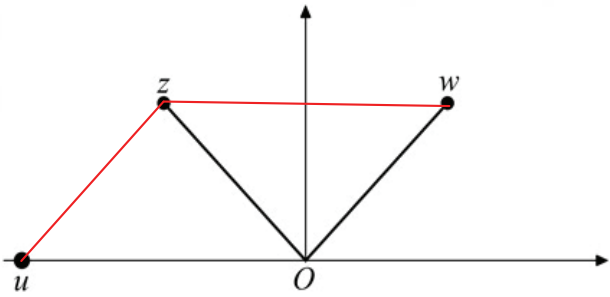
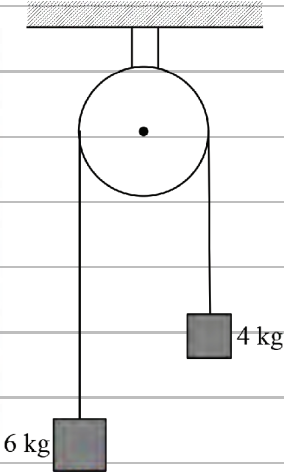
(You may assume that $\frac{z}{1-z} = \frac{i}{2\sin \frac{\theta}{2}} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$.)

End of Examination

Year 12 Mathematics Extension 2 Trial HSC Solutions 2025

Question	Suggested solution	Marking guidelines	Markers Notes
1.	'for any integers m and n, if $m+n \geq 9$ then $m \geq 5$ or $n \geq 5$ Contrapositive $\therefore$ switch the hypothesis and the conclusion and negate both $\therefore$ If $m < 5$ and $n < 5$, then $m+n < 9$ (C)		
2.	 amplitude = 2 period = $\frac{2\pi}{n} = 2$ $\frac{2\pi}{2} = n$ $n = \pi$ $\therefore x = a \cos(nt)$ $= 2 \cos(\pi t)$ (A)		
3.	$z = -(2a+1) + 2ai$ $\therefore \bar{z} = -(2a+1) - 2ai$ $\frac{4a}{1+\bar{z}} = \frac{4a}{1+-(2a+1)-2ai}$ $= \frac{\cancel{4a}}{-2\cancel{a} - 2\cancel{a}i} \times \frac{-2\cancel{a} + 2\cancel{a}i}{-2\cancel{a} + 2\cancel{a}i}$ $= \frac{4(-2+2i)}{(-2)^2 - (2i)^2}$ $= \frac{-8+8i}{4+4}$ $= \frac{-8+8i}{8}$ $= -1+i$ $= \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$ $-1+i = \sqrt{1^2+1^2}$ $= \sqrt{2}$ $\arg(-1+i) = \tan^{-1}\left(\frac{1}{-1}\right)$ $= \frac{3\pi}{4}$ (B)		

Question	Suggested solution	Marking guidelines	Markers Notes
4.	 $\frac{\vec{AE}}{\vec{AB}} = \frac{7}{4}$$\therefore \vec{AE} = \frac{7}{4} \vec{a}$ $\vec{FA} = \vec{FD} + \vec{DA}$$= -\frac{1}{2}\vec{a} - \vec{b}$ $\vec{FE} = \vec{FA} + \vec{AE}$$= \frac{7}{4}\vec{a} - \frac{1}{2}\vec{a} - \vec{b}$$= \frac{5}{4}\vec{a} - \vec{b}$ A		
5.	$u = 1 + e^x \quad x = \ln 2 \quad u = 1 + e^{\ln 2}$$\frac{du}{dx} = e^x \quad = 1 + 2$$\frac{du}{e^x} = dx \quad x = 0 \quad u = 1 + e^0$$= 1 + 1$$= 2$ $\therefore \int_0^{\ln 2} \frac{1}{1+e^x} dx = \int_2^3 \frac{1}{u} \cdot \frac{du}{e^x}$$= \int_2^3 \frac{1}{u(u-1)} du$$= \int_2^3 \left(\frac{1}{u-1} - \frac{1}{u} \right) du$ D	$\frac{1}{u(u-1)} = \frac{A}{u} + \frac{B}{u-1}$$1 = A(u-1) + Bu$$u=1 \quad 1 = B$$u=0 \quad 1 = A(-1)$$A = -1$	

Question	Suggested solution	Marking guidelines	Markers Notes
6.	 $\arg(u) = \arg(z) + \arg(w)$ $= \arg(zw)$ $\therefore u = zw$ $\therefore \text{Either A or B}$ $w + u = z \quad (\text{parallelogram method})$ $\therefore u = z - w$ (B)		
7.	Let: $m_1 = 6 \text{ kg}$ (heavier mass - accelerates downward) $m_2 = 4 \text{ kg}$ (lighter mass - accelerates upwards) T = tension in string a = acceleration 		

Question	Suggested solution	Marking guidelines	Markers Notes
	For 6 kg mass: $m_1 g - T = m_1 a \quad (1)$ For 4 kg mass $T - m_2 g = m_2 a \quad (2)$ $(1) + (2)$ $m_1 g - m_2 g = m_1 a + m_2 a$ $g(m_1 - m_2) = a(m_1 + m_2)$ $g(6 - 4) = a(6 + 4)$ $2g = 10a$ $a = \frac{2g}{10}$ $= \frac{g}{5}$ (A) $1 + \sqrt{2} > \sqrt{5}$ $\therefore E$ is the greatest distance away from i on the locus of z (D)		$m_1 g - m_1 a = T$ $T = m_2 a + m_2 g$

Question	Suggested solution	Marking guidelines	Markers Notes
9.	Line segment 1: $\underline{r} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \end{pmatrix}, \quad -2 \leq \lambda \leq 2$ • Direction vector: $\underline{d}_1 = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ • Length: $\lambda \leq 2 \Rightarrow \text{Total length} = 4 \times \underline{d}_1$ $\underline{d}_1 = \sqrt{(-3)^2 + 2^2}$ $= \sqrt{9+4}$ $= \sqrt{13}$ $\therefore \text{length} = 4\sqrt{13}$ Line segment 2: $\underline{r} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad 0 \leq \mu \leq 2$ • Direction vector: $\underline{d}_2 = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$ • Length: $2 \times \underline{d}_2$ $= 2 \times \sqrt{4^2 + 6^2}$ $= 2 \times \sqrt{16 + 36}$ $= 2\sqrt{52}$ $= 4\sqrt{13}$ Line segment 3: $\underline{r} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}, \quad 1 \leq \lambda \leq 3$ • Direction vector: $\underline{d}_3 = \begin{pmatrix} 9 \\ -6 \end{pmatrix} = -3 \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ • Length: $(3-1) \times \underline{d}_3 = 2 \times \sqrt{9^2 + (-6)^2}$ $= 2 \times \sqrt{81 + 36}$ $= 2\sqrt{117}$		

Question	Suggested solution	Marking guidelines	Markers Notes
	$\therefore \vec{r} \parallel \vec{v}, \vec{v} = \vec{r} ,$ $\vec{d}_1 \cdot \vec{d}_2 = -3 \times 4 + 2 \times 6 = 0 \quad \text{or} \quad \vec{d}_2 \cdot \vec{d}_3 = 4 \times 9 + (-6) \times 6 = 0$ $\therefore \vec{v} \perp \vec{r}$ $\therefore \textcircled{C} \text{ is not true}$		
10.	$\frac{x}{2 \cos x} \text{ and } \tan^{-1}(x^3) \text{ are}$ odd functions, $\therefore$ definite integrals involving limits indicated will be 0. $x^3 \sin x$ and $e^{-x^2} - 1$ are even functions, $\therefore$ definite integrals involving indicated limits will be non-zero. $e^{-x^2} - 1 \leq 0$ in the domain of the integral $\therefore$ value of integral will be < 0 $x^3 \sin x > 0$ in the domain of the integral $\therefore$ value of integral will be > 0 $\textcircled{B}$		

Question	Suggested solution	Marking guidelines	Markers Notes
11. a)	(i) $\vec{AB} = \vec{AO} + \vec{OB}$ $\vec{AC} = \vec{AO} + \vec{OC}$ $= \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} \quad = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} \quad = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$ (ii) $\vec{AB} \cdot \vec{AC} = 2(-2) + (-2)(2) + 3(2)$ $= -2$ $ \vec{AB} = \sqrt{2^2 + (-2)^2 + 3^2} \quad \vec{AC} = \sqrt{(-2)^2 + 2^2 + 2^2}$ $= \sqrt{17} \quad = \sqrt{12}$ $\cos \theta = \frac{\vec{AB} \cdot \vec{AC}}{ \vec{AB} \vec{AC} }$ $= \frac{-2}{\sqrt{17} \times \sqrt{12}}$ $= -0.14 \dots$ $\theta = \cos^{-1}(-0.14 \dots)$ $= 98.04 \dots$ $= 98^\circ$ $\therefore$ acute $\angle$ between $\vec{AB}$ and $\vec{AC}$ is 92°	1 1 1 1	
b)	$\int \frac{1}{x^2 + 2x + 3} dx = \int \frac{1}{(x+1)^2 + 2} dx$ $= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + C$	1 1	

Question	Suggested solution	Marking guidelines	Markers Notes
11 c)	$\arg[(b-i)^3] = -\frac{\pi}{2}$ $\therefore \arg(b-i) = -\frac{\pi}{2} \div 3$ $= -\frac{\pi}{6}$ $\therefore \tan\left(-\frac{\pi}{6}\right) = -\frac{1}{b}$ $-\frac{1}{\sqrt{3}} = -\frac{1}{b}$ $\therefore b = \sqrt{3}$ OR $(b-i)^3 = b^3 - 3b^2i + 3bi^2 - i^3$ $= b^3 - 3b^2i - 3b + i$ $= b^3 - 3b + (1 - 3b^2)i$ If $\arg(z) = -\frac{\pi}{2}$ then $b^3 - 3b = 0$ $b(b^2 - 3) = 0$ $\therefore b = \sqrt{3} \text{ since } b > 0$	① ① ① ① ①	
d)	Let $z = x+iy$ Then $(2-3i)(x+iy) = (2x+3y) + (2y-3x)i$ $6 \leq \operatorname{Re}[(2-3i)z] < 12 \Rightarrow 6 \leq 2x+3y < 12$ and $\operatorname{Re} z \times \operatorname{Im} z \geq 0 \Rightarrow xy \geq 0$	② - correct region provided ① - 1 error ① - more than 1 error	

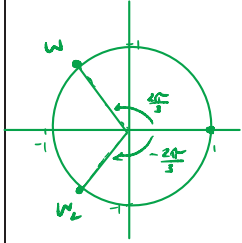
Question	Suggested solution	Marking guidelines	Markers Notes
11 e)	(i) $\frac{x}{(x-2)^2(x-1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-1)}$ $\therefore x \equiv A(x-1)(x-2) + B(x-1) + C(x-2)^2$ $= A(x^2 - 3x + 2) + B(x-1) + C(x^2 - 4x + 4)$ $= (A+C)x^2 + (B-3A-4C)x + 2A - B + 4C$ Let: $x=1, 1 = C(-1)^2$ $\therefore C = 1$ $x=2, 2 = B(2-1)$ $\therefore B = 2$ Equating coefficients of x^2: $A+C = 0$ $A+1 = 0$ $A = -1$ $\therefore \frac{x}{(x-2)^2(x-1)} = \frac{-1}{x-2} + \frac{2}{(x-2)^2} + \frac{1}{x-1}$ (ii) $\int_3^4 \frac{x}{(x-2)^2(x-1)} dx = \int_3^4 \left[\frac{-1}{x-2} + \frac{2}{(x-2)^2} + \frac{1}{x-1} \right] dx$ $= \left[-\ln x-2 - \frac{2}{x-2} + \ln x-1 \right]_3^4$ $= \left[\ln \left \frac{x-1}{x-2} \right - \frac{2}{x-2} \right]_3^4$ $= \ln \left \frac{4-1}{4-2} \right - \frac{2}{4-2} - \left(\ln \left \frac{3-1}{3-2} \right - \frac{2}{3-2} \right)$ $= \ln \left(\frac{3}{2} \right) - 1 - \ln(2) + 2$ $= \ln \left(\frac{3}{4} \right) + 1$	① ① ① ①	

Question	Suggested solution	Marking guidelines	Markers Notes	
12. a)	Let $n = 2k$, where $k \in \mathbb{Z}$ Then $n^3 - n = (2k)^3 - 2k$ $= 8k^3 - 2k$ $= 2k(4k^2 - 1), \text{ which is divisible by } 2$ To show divisibility by 6, also need to show it's divisible by 3. $= 2k(2k-1)(2k+1)$ $= (2k-1)2k(2k+1)$ This is a product of 3 consecutive integers: $2k-1, 2k, 2k+1$ Among any three consecutive integers, one is divisible by 3 $\therefore$ Expression is also divisible by 3 Hence divisible by $\text{LCM}(2, 3) = 6$	① ①		
b)	(i) $\frac{d}{dx}(\sec \theta) = \frac{d}{dx}\left(\frac{1}{\cos \theta}\right)$ $= \frac{d}{dx}[(\cos \theta)^{-1}]$ $= -1(\cos \theta)^{-2} \times (-\sin \theta)$ $= \frac{\sin \theta}{\cos^2 \theta}$ $= \frac{\sin \theta}{\cos \theta} \times \frac{1}{\cos \theta}$ $= \tan \theta \sec \theta \text{ as required}$	①		
	(ii) $x = \sec \theta - 1$ $\frac{dx}{d\theta} = \sec \theta \tan \theta$ $dx = \sec \theta \tan \theta d\theta$	$x = 1 \rightarrow 1 = \sec \theta - 1$ $2 = \sec \theta$ $\frac{1}{2} = \cos \theta$ $\theta = \frac{\pi}{3}$ $x = \sqrt{2} - 1 \rightarrow \sqrt{2} - 1 = \sec \theta$ $\frac{1}{\sqrt{2}} = \cos \theta$ $\theta = \frac{\pi}{4}$	①	

Question	Suggested solution	Marking guidelines	Markers Notes
	$\int_{\sqrt{2}-1}^1 \frac{2}{(x+1)\sqrt{x^2+2x}} dx$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 \cancel{\sec \theta} \tan \theta d\theta}{\cancel{\sec \theta} \sqrt{(\sec \theta - 1)^2 + 2(\sec \theta - 1)}}$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 \tan \theta d\theta}{\sqrt{\sec^2 \theta - 2\cancel{\sec \theta} + 1 + 2\cancel{\sec \theta} - 2}}$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 \tan \theta d\theta}{\sqrt{\tan^2 \theta}}$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 2 d\theta$ $= 2 \left[\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$ $= 2 \left[\frac{\pi}{3} - \frac{\pi}{4} \right]$ $= \frac{\pi}{6}$	① ① ①	
12 c)	Let $\underline{w} = x\underline{i} + y\underline{j} + z\underline{k}$ be a unit vector. such that: $\underline{u} \cdot \underline{w} = -5x + 2y + 3z = 0 \quad \textcircled{1}$ $\underline{v} \cdot \underline{w} = 2x - y - 2z = 0 \quad \textcircled{2}$ and $x^2 + y^2 + z^2 = 1 \quad \textcircled{3}$		

Question	Suggested solution	Marking guidelines	Markers Notes
	$\textcircled{1} + 2 \times \textcircled{2} : \quad -x - z = 0$ $\therefore z = -x \quad \textcircled{4}$		
	$\textcircled{1} \times 2 + 3 \times \textcircled{2} : \quad -4x + y = 0$ $y = 4x \quad \textcircled{5}$	①	
	sub into $\textcircled{3}$		
	$x^2 + (4x)^2 + (-x)^2 = 1$ $x^2 + 16x^2 + x^2 = 1$ $18x^2 = 1$ $x^2 = \frac{1}{18}$ $x = \pm \frac{1}{3\sqrt{2}}$	①	
	$y = \pm \frac{4}{3\sqrt{2}}$		
	$z = \mp \frac{1}{3\sqrt{2}}$		
	$\therefore \frac{1}{3\sqrt{2}} (\underset{\sim}{i} + 4\underset{\sim}{j} - \underset{\sim}{k}) \text{ is a vector}$ perpendicular to both $\underset{\sim}{u}$ and $\underset{\sim}{v}$	①	

Question	Suggested solution	Marking guidelines	Markers Notes
12 d) (i)	$\vec{PQ} = \vec{OQ} - \vec{OP}$ $= \begin{pmatrix} -1 \\ 12 \\ -2 \end{pmatrix} - \begin{pmatrix} -2 \\ 14 \\ -5 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \Rightarrow \text{direction vector}$ $\therefore \vec{r}_1 = \begin{pmatrix} -2 \\ 14 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ $= (-2+\lambda)\underline{i} + (14-2\lambda)\underline{j} + (-5+3\lambda)\underline{k}$	① ①	
(ii)	$-2+\lambda = 2+a\mu \quad \text{①}$ $14-2\lambda = 27+(a+1)\mu \quad \text{②}$ $-5+3\lambda = 1+(a+2)\mu \quad \text{③}$	①	
	From ① $a\mu = \lambda - 4$		
	From ② $14-2\lambda = 27+\lambda-4+\mu$ $-9 = 3\lambda + \mu \quad \text{⑤}$		
	From ③ $-5+3\lambda = 1+\lambda-4+2\mu$ $2\lambda - 2\mu = 2$ $\lambda - \mu = 1 \quad \text{⑥}$		
	⑤+⑥ $4\lambda = -8$ $\lambda = -2$	①	
	$\therefore \vec{r}_1 = (-2-2)\underline{i} + (14+4)\underline{j} + (-5-6)\underline{k}$ $= -4\underline{i} + 18\underline{j} - 11\underline{k}$ $\therefore R = (-4, 18, -11)$	①	

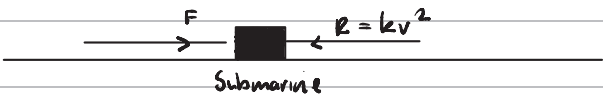
Question	Suggested solution	Marking guidelines	Markers Notes
13. a)	$w^3 = 1 \quad \text{but } w \neq 1$ $\therefore w^3 - 1 = 0$ $(w-1)(w^2 + w + 1) = 0$ $\therefore 1 + w + w^2 = 0 \quad \text{since } w \neq 1$ $\sum_{k=0}^{1001} w^k = 1 + w + w^2 + \dots + w^{1001}$ In this sum there are 1002 terms, which is divisible by 3 $1002 = 334$ $\therefore \sum_{k=0}^{1001} w^k = (1 + w + w^2) + (w^3 + w^4 + w^5) + \dots + (w^{996} + w^{997} + w^{998}) + (w^{999} + w^{1000} + w^{1001})$ $= (1 + w + w^2) + w^3(1 + w + w^2) + \dots + w^{996}(1 + w + w^2) + w^{999}(1 + w + w^2)$ $= 0 + 0 + \dots + 0 + 0$ $= 334 \times 0$ $= 0$	① ①	
b) (i)	$I_n = \int_0^{\pi/4} \tan^n x \, dx$ $= \int_0^{\pi/4} \tan^{n-2} x \tan^2 x \, dx$ $= \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x - 1) \, dx$ $= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx - \int_0^{\pi/4} \tan^{n-2} x \, dx$ $= \frac{1}{n-1} \left[\tan^{n-1} x \right]_0^{\pi/4} - I_{n-2}$ $= \frac{1}{n-1} \left[\tan^{n-1} \frac{\pi}{4} - \tan^{n-1} 0 \right] - I_{n-2}$		$\int \sec^2 x [\tan x]^n \, dx$ $= \int f'(x) [f(x)]^n \, dx$

Question	Suggested solution	Marking guidelines	Markers Notes
13 (c)	$\ddot{x} = x-1 \quad t=0, x=0, v=1$		
	(i) $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = x-1$	①	
	$\frac{1}{2}v^2 = \frac{x^2}{2} - x + C_1$		
	when $x=0 \quad v=1$		
	$\frac{1}{2} = 0 - 0 + C_1$		
	$\frac{1}{2} = C_1$		
	$\therefore \frac{1}{2}v^2 = \frac{x^2}{2} - x + \frac{1}{2}$		
	$v^2 = x^2 - 2x + 1$		
	$v = \pm \sqrt{x^2 - 2x + 1}$		
	$= \pm \sqrt{(x-1)^2}$	①	
	$= \pm (x-1)$		
	But when $x=0 \quad v=1$		
	$\therefore v = -(x-1)$		
	$= 1-x \quad \text{as required}$	①	
	(ii) $v = 1-x$		
	$\frac{dx}{dt} = 1-x$		
	$\frac{dt}{dx} = \frac{1}{1-x}$	①	
	$t = -\ln 1-x + C_2$		
	when $t=0 \quad x=0$		
	$0 = -\ln 1-0 + C_2$		
	$= C_2$		
	$\therefore t = -\ln 1-x $		
	$-t = \ln 1-x $		
	$\therefore e^{-t} = 1-x$		
	$x = 1 - e^{-t}$	①	

Question	Suggested solution	Marking guidelines	Markers Notes
13 c)	(iii) $x = 1 - e^{-t}$ As $t \rightarrow \infty$, $e^{-t} \rightarrow 0$ $\Rightarrow x \rightarrow 1 - 0$ $= 1$ $\therefore$ limiting position is $x = 1$	①	

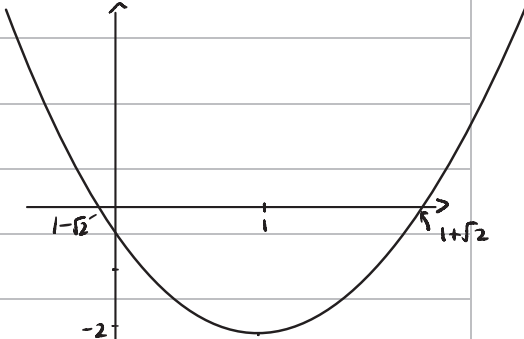
Question	Suggested solution	Marking guidelines	Markers Notes
14 a)	(i) Prove $\frac{x+y}{2} \geq \sqrt{xy}$ Consider $\frac{x+y}{2} - \sqrt{xy} \geq 0$ LHS = $\frac{x+y}{2} - \sqrt{xy}$ = $\frac{x - 2\sqrt{xy} + y}{2}$ = $\frac{(\sqrt{x} - \sqrt{y})^2}{2}$ ≥ 0 since $(\sqrt{x} - \sqrt{y})^2 > 0$ if $x \neq y$ and $(\sqrt{x} - \sqrt{y})^2 = 0$ if $x = y$ $\therefore \frac{x+y}{2} \geq \sqrt{xy}$ (ii) using result from part (i) $\frac{a^4+b^4}{2} \geq \sqrt{a^4b^4} = a^2b^2$ ① $\frac{b^4+c^4}{2} \geq \sqrt{b^4c^4} = b^2c^2$ ② $\frac{a^4+c^4}{2} \geq \sqrt{a^4c^4} = a^2c^2$ ③ ① + ② + ③ : $\frac{(a^4+b^4) + (b^4+c^4) + (a^4+c^4)}{2} \geq a^2b^2 + b^2c^2 + a^2c^2$ $\cancel{2}(a^4+b^4+c^4) \geq a^2b^2 + b^2c^2 + a^2c^2$ $\therefore a^4+b^4+c^4 \geq a^2b^2 + b^2c^2 + a^2c^2$	① ① ①	

Question	Suggested solution	Marking guidelines	Markers Notes
14 a)	(iii) Using the result in part (i) again, then $\frac{a^2b^2 + a^2c^2}{2} = \frac{a^2(b^2+c^2)}{2} \geq \sqrt{a^2b^2 \cdot a^2c^2} = a^2bc$ $\frac{b^2c^2 + b^2a^2}{2} = \frac{b^2(c^2+a^2)}{2} \geq \sqrt{b^2c^2 \cdot b^2a^2} = b^2ca$ $\frac{c^2a^2 + c^2b^2}{2} = \frac{c^2(a^2+b^2)}{2} \geq \sqrt{c^2a^2 \cdot c^2b^2} = c^2ab$ ① + ② + ③ : $\frac{a^2(b^2+c^2) + b^2(c^2+a^2) + c^2(a^2+b^2)}{2} \geq a^2bc + b^2ca + c^2ab$ i.e. $\frac{a^2b^2 + b^2c^2 + c^2a^2}{2}$ $\geq a^2bc + b^2ca + c^2ab$ $\therefore a^2b^2 + b^2c^2 + c^2a^2 \geq a^2bc + b^2ca + c^2ab$ (iv) From (ii) and (iii) $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + a^2c^2 \quad \text{and}$ $a^2b^2 + b^2c^2 + c^2a^2 \geq a^2bc + b^2ca + c^2ab$ $\therefore a^4 + b^4 + c^4 \geq a^2bc + b^2ca + c^2ab$ $= abc(a+b+c)$ $= abcd \quad \text{since } a+b+c=d$	① ② ③ ① ①	①

Question	Suggested solution	Marking guidelines	Markers Notes
14 b)	(i)  Resultant force in $\rightarrow$ direction is $F - R = F - kv^2$ By Newton's Law of Motion, if the acceleration of submarine (of mass m) in $\rightarrow$ direction is 'a' where $a = \frac{dv}{dt}$, then the force acting on the submarine in $\rightarrow$ direction is $ma = m \cdot \frac{dv}{dt}$ Thus $m \cdot \frac{dv}{dt} = F - kv^2$ $\therefore \frac{dv}{dt} = \frac{1}{m} (F - kv^2)$ (ii) $\frac{dv}{dt} = v \frac{dv}{dx}$ $\therefore v \frac{dv}{dx} = \frac{1}{m} (F - kv^2)$ $\frac{dv}{dx} = \frac{1}{m} \cdot \frac{(F - kv^2)}{v}$ $\frac{dx}{dv} = m \cdot \frac{v}{F - kv^2}$ $x = m \int \frac{v}{F - kv^2} dv$ The distance x travelled during the period when submarine increases its speed from v_1 to v_2 is given by	① ①	

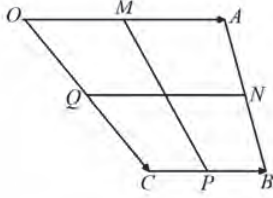
Question	Suggested solution	Marking guidelines	Markers Notes
	$X = \frac{m}{-2k} \int_{v_1}^{v_2} \frac{-2k v dv}{F - kv^2}$ $= -\frac{m}{2k} \left[\ln F - kv^2 \right]_{v_1}^{v_2}$ $= -\frac{m}{2k} \left[\ln F - kv_2^2 - \ln F - kv_1^2 \right]$ $= -\frac{m}{2k} \log_e \left \frac{F - kv_2^2}{F - kv_1^2} \right $ $= \frac{m}{2k} \log_e \left \frac{F - kv_1^2}{F - kv_2^2} \right $	① ①	
c)	(i) Prove $1 - x \leq e^{-x}$ for $x \geq 0$ Consider $1 - x - e^{-x} \leq 0$ Let $f(x) = 1 - x - e^{-x}$ then $f'(x) = -1 + e^{-x}$ ≤ 0 for $x \geq 0$ $\therefore f(x)$ is decreasing or stationary as x increases for $x \geq 0$ Also $f(0) = 0$ $\therefore f(x) \leq 0$ for $x \geq 0$ Hence $1 - x - e^{-x} \leq 0$ $\Rightarrow 1 - x \leq e^{-x}$	① ①	
	(ii) $P(T=0) = P(X_k=0 \forall k=1,2,\dots,n)$ $= (1-p_1)(1-p_2)\dots(1-p_n)$ since the X_k are independent Bernoulli random variables taking the value 1 with probability p_k and 0 with probability $1 - p_k$	①	

Question	Suggested solution	Marking guidelines	Markers Notes
14 c)	Then using part (i), $p_k \geq 0 \forall k$ $\Rightarrow P(T=0) \leq e^{-p_1} e^{-p_2} \dots e^{-p_n}$ $= e^{-(p_1+p_2+\dots+p_n)}$ But $\mu = \sum_{k=1}^n E(X_k)$ $= \sum_{k=1}^n p_k$ $\therefore P(T=0) \leq e^{-\mu}$	①	

Question	Suggested solution	Marking guidelines	Markers Notes
15. a)	(i) $x^2 > 2x + 1$ $x^2 - 2x - 1 > 0$ $(x-1)^2 - 2 > 0$ $(x-1-\sqrt{2})(x-1+\sqrt{2}) > 0$  $\therefore x^2 - 2x - 1 > 0$ for $x < 1-\sqrt{2}$ or $x > 1+\sqrt{2}$ i.e. $x^2 > 2x + 1$ for $x < 1-\sqrt{2}$ or $x > 1+\sqrt{2}$ (ii) Prove $2^n > n^2$ for $n \geq 5$ Step 1 Prove true for $n=5$ LHS = 2^5 RHS = 5^2 = 32 = 25 LHS > RHS $\therefore$ True for $n=5$ Step 2: Assume true for $n=k$, $k \geq 5$ i.e. $2^k > k^2$ $\therefore 2^k - k^2 > 0$ Step 3: Prove true for $n=k+1$ i.e. $2^{k+1} - (k+1)^2 > 0$ LHS = $2^{k+1} - (k+1)^2$ = $2 \cdot 2^k - k^2 - 2k - 1$ = $2 \cdot 2^k - 2k^2 + k^2 - 2k - 1$ = $2(2^k - k^2) + (k^2 - 2k - 1)$ > 0	① ①	

Question	Suggested solution	Marking guidelines	Markers Notes
	Since for $k \geq 5$: $2^2 - k^2 > 0$ using assumption and from (i) $k^2 - 2k - 1 > 0$ for $k > 1 + \sqrt{2} \approx 2.4$ $\therefore$ Also true for $k \geq 5$ Step 4: Conclusion True for $n = k+1$ if true for $n = k$ True for $n = 5$ $\therefore$ by induction true for all integers $n \geq 5$. b) Assume that there exists a rational number $x \in \mathbb{Q}$ such that $x^2 = 2$ Since $x \in \mathbb{Q}$, $x = \frac{p}{q}$, where $p, q \in \mathbb{Z}, q \neq 0$ and p, q have no common factors $\therefore \left(\frac{p}{q}\right)^2 = 2$ $\frac{p^2}{q^2} = 2$ $p^2 = 2q^2$ $\therefore p^2$ is even $\Rightarrow p$ must be even (since the square of an odd number is odd) Let $p = 2k, k \in \mathbb{Z}$ $(2k)^2 = 2q^2$ $4k^2 = 2q^2$ $2k^2 = q^2$ $\therefore q^2$ is even $\Rightarrow q$ is also even $\therefore p$ and q are both even meaning they have a common factor of 2 This contradicts the assumption that $\frac{p}{q}$ is in simplest form $\therefore$ by contradiction if $x \in \mathbb{Q}, x^2 \neq 2$	① ① ①	

Question	Suggested solution	Marking guidelines	Markers Notes
	(iii) Noting $i^2 = -1$, then the complex roots of $P(z) = 0$ are those when $(\sqrt{2}z^2 + 1)(z^4 + \sqrt{2}) = 0$ i.e. $(\sqrt{2}z^2 - i^2)(z^2 - \sqrt{2}i^2) = 0$ i.e. $(2^{\frac{1}{4}}z - i)(2^{\frac{1}{4}}z + i)(z - 2^{\frac{1}{4}}i)(z + 2^{\frac{1}{4}}i) = 0$ i.e. $z = i2^{-\frac{1}{4}}, -i2^{-\frac{1}{4}}, i2^{\frac{1}{4}}, -i2^{\frac{1}{4}}$ $\therefore$ All the roots of $P(z) = 0$ are $z = \pm 2^{-\frac{1}{4}}, \pm 2^{\frac{1}{4}}, \pm i2^{-\frac{1}{4}}, \pm i2^{\frac{1}{4}}$ Note: If $w = i2^{\frac{1}{4}}$, then $iw = -2^{\frac{1}{4}}, \quad \frac{1}{w} = \frac{2^{-\frac{1}{4}}}{i} = -i2^{-\frac{1}{4}}$ which are also roots of $P(z) = 0$, from part (i) d) Let $\underline{p} = (a, b, c)$, $\underline{q} = (e, f, g)$ and $\underline{r} = (x, y, z)$ $\begin{aligned} \text{LHS} &= \underline{p} \cdot (\underline{q} + \underline{r}) \\ &= (a, b, c) \cdot (e+x, f+y, g+z) \\ &= a(e+x) + b(f+y) + c(g+z) \\ &= ae + ax + bf + by + cg + cz \end{aligned}$ $\begin{aligned} \text{RHS} &= \underline{p} \cdot \underline{q} + \underline{p} \cdot \underline{r} \\ &= (a, b, c) \cdot (e, f, g) + (a, b, c) \cdot (x, y, z) \\ &= ae + bf + cg + ax + by + cz \\ &= ae + ax + bf + by + cg + cz \\ &= \text{LHS} \end{aligned}$ $\therefore \underline{p} \cdot (\underline{q} + \underline{r}) = \underline{p} \cdot \underline{q} + \underline{p} \cdot \underline{r}$	① ① 3m - correctly worked proof with conclusion 2m - for worked proof with minor error in logic or algebraic manipulation 1m - for some relevant algebraic working	

Question	Suggested solution	Marking guidelines	Markers Notes
16 b)	 (i) N is the midpoint of AB $\therefore \vec{ON} = \frac{1}{2}(\vec{OA} + \vec{OB})$ $\therefore \vec{ON} = \frac{1}{2}(\underline{a} + (\underline{c} + k\underline{a}))$ $= \frac{1}{2}\underline{c} + \frac{1}{2}(k+1)\underline{a}$ $\therefore \frac{1}{2}(\vec{OQ} + \vec{ON}) = \frac{1}{2}(\frac{1}{2}\underline{c} + \frac{1}{2}\underline{c} + \frac{1}{2}(k+1)\underline{a})$ $= \frac{1}{2}\underline{c} + \frac{1}{4}(k+1)\underline{a}$ But $\frac{1}{2}(\vec{OM} + \vec{OP}) = \frac{1}{2}(\frac{1}{2}\underline{a} + (\underline{c} + \frac{1}{2}k\underline{a}))$ $= \frac{1}{2}\underline{c} + \frac{1}{4}(k+1)\underline{a}$ Hence QN and MP have a common midpoint T where $\vec{OT} = \frac{1}{2}\underline{c} + \frac{1}{4}(k+1)\underline{a}$ Hence QN and MP bisect each other in their intersection point T. (ii) $\vec{MP} = \vec{MO} + \vec{OC} + \vec{CP}$ $= -\frac{1}{2}\underline{a} + \underline{c} + \frac{1}{2}k\underline{a}$ $= \underline{c} + \frac{1}{2}(k-1)\underline{a}$ $\vec{QN} = \vec{QO} + \vec{ON}$ $= -\frac{1}{2}\underline{c} + \frac{1}{2}\underline{c} + \frac{1}{2}(k+1)\underline{a}$ $= \frac{1}{2}(k+1)\underline{a}$ using expression for $\vec{ON}$ from (i)	① ① ①	since diagonals of a parallelogram bisect each other

Question	Suggested solution	Marking guidelines	Markers Notes
16 b)	(ii) Hence if $MP \perp QN$		
	$\left(\underline{c} + \frac{1}{2}(k-1)\underline{a}\right) \cdot \left(\frac{1}{2}(k+1)\underline{a}\right) = 0$		
	$\frac{1}{2}(k+1)\underline{a} \cdot \underline{c} + \frac{1}{4}(k-1)(k+1)\underline{a} \cdot \underline{a} = 0$		
	$\frac{1}{2}(k+1)\cancel{\underline{a}} \cdot \underline{c} = -\frac{1}{4}(k-1)(k+1)\cancel{\underline{a}} \cdot \underline{a}$		
	$\underline{a} \cdot \underline{c} = -\frac{1}{2}(k-1)\underline{a} \cdot \underline{a}$	①	
	$\vec{AB} = \vec{AO} + \vec{OC} + \vec{CB}$		
	$= -\underline{a} + \underline{c} + k\underline{a}$		
	$= \underline{c} + (k-1)\underline{a}$	①	
	$\therefore \vec{AB} ^2 = (\underline{c} + (k-1)\underline{a}) \cdot (\underline{c} + (k-1)\underline{a})$		
	$= \underline{c} \cdot \underline{c} + 2(k-1)\underline{a} \cdot \underline{c} + (k-1)^2 \underline{a} \cdot \underline{a}$		
	$= \underline{c} \cdot \underline{c} + (k-1)^2 \underline{a} \cdot \underline{a} + 2(k-1)\left(-\frac{1}{2}(k-1)\right)\underline{a} \cdot \underline{a}$		if $MP \perp QN$
	$= \underline{c} \cdot \underline{c} + \{(k-1)^2 - (k-1)^2\}\underline{a} \cdot \underline{a}$		
	$= \underline{c} \cdot \underline{c}$		
	$= \vec{OC} ^2$	①	
c)	(i) $z + z^2 + \dots + z^n$ is a geometric series		
	with $a=z$ and $r=z \quad z \neq 1$		
	$\therefore S_n = \frac{z(1-z^{n+1})}{1-z}$		
	$\therefore \sum_{k=1}^n (z + z^2 + \dots + z^k) = \sum_{k=1}^n \frac{z(1-z^{k+1})}{1-z}$		
	$= \frac{z}{1-z} \sum_{k=1}^n (1-z^{k+1})$		
	$= \frac{z}{1-z} [(1-z) + (1-z^2) + (1-z^3) + \dots + (1-z^{n+1})]$		
	$= \frac{z}{1-z} [n + 1 - (z + z^2 + z^3 + \dots + z^{n+1})]$		
	$= \frac{z}{1-z} \left[n + 1 - \frac{z(1-z^{n+1})}{1-z} \right]$		
	$= \frac{n+1}{1-z} - \frac{z^2(1-z^{n+1})}{(1-z)^2}, \quad z \neq 1$		Note: $\frac{z}{1-z}$ is independent of k

Question	Suggested solution	Marking guidelines	Markers Notes
16 b)	(ii) If $z = \cos \theta + i \sin \theta$, then by de Moivre $z^2 = \cos 2\theta + i \sin 2\theta$ $z^3 = \cos 3\theta + i \sin 3\theta \text{ and so on}$ Now $\sum_{k=1}^n (z + z^2 + z^3 + \dots + z^k)$ $= \sum_{k=1}^n (\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos k\theta)$ $= \sum_{k=1}^n [\cos \theta + \cos 2\theta + \dots + \cos k\theta + i(\sin \theta + \sin 2\theta + \dots + \sin k\theta)]$ Now $\text{Im}(z + z^2 + \dots + z^k)$ $= \sin \theta + \sin 2\theta + \dots + \sin k\theta$ $\therefore \text{Im} \sum_{k=1}^n (z + z^2 + \dots + z^k) = \sum_{k=1}^n (\sin \theta + \sin 2\theta + \dots + \sin k\theta) \quad (1)$ Given that $\frac{z}{1-z} = \frac{i}{2\sin \frac{\theta}{2}} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$ $\therefore \text{Im} \frac{nz}{1-z} = \text{Im} \frac{n}{2\sin \frac{\theta}{2}} (i \cos \frac{\theta}{2} - \sin \frac{\theta}{2})$ $= \frac{n}{2\sin \frac{\theta}{2}} \times \cos \frac{\theta}{2} \quad (1)$ Also $\frac{z^2}{(1-z)^2} (1-z^n) = \left[\frac{i}{2\sin \frac{\theta}{2}} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}) \right]^2 \times [1 - (\cos \theta + i \sin \theta)^n]$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} (\cos \theta + i \sin \theta) [(1 - \cos n\theta) - i \sin(n\theta)]$ $\therefore \text{Im} \frac{z^2}{(1-z)^2} (1-z^n)$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} \times \text{Im} \{ (\cos \theta + i \sin \theta) [(1 - \cos n\theta) - i \sin(n\theta)] \}$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} \{ \cos \theta (-\sin(n\theta)) + \sin \theta (1 - \cos n\theta) \}$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} \{ \sin \theta - (\sin(n\theta) \cos \theta + \sin \theta \cos(n\theta)) \}$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} \{ \sin \theta - \sin(n\theta + \theta) \}$ $= \frac{-1}{4 \sin^2 \frac{\theta}{2}} \{ \sin \theta - \sin(n+1)\theta \} \quad (1)$		

Question	Suggested solution	Marking guidelines	Markers Notes
	$\therefore \operatorname{Im} \left\{ \frac{nz}{1-z} - \frac{z^2}{(1-z)^2} (1-z^2) \right\}$ $= \frac{n \cos \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} + \frac{1}{4 \sin^2 \frac{\theta}{2}} \{ \sin \theta - \sin(n+1)\theta \}$ $= \frac{2n \sin \frac{\theta}{2} \cos \frac{\theta}{2} + \sin \theta - \sin(n+1)\theta}{4 \sin^2 \frac{\theta}{2}}$ $= \frac{n \sin \theta + \sin \theta - \sin(n+1)\theta}{4 \sin^2 \frac{\theta}{2}}$ $= \frac{(n+1) \sin \theta - \sin(n+1)\theta}{4 \sin^2 \frac{\theta}{2}}$	①	